UMUT ÖZDİL

MACHINE LEARNING ENGINEER

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Machine Learning Engineer with a solid foundation in software engineering. Experienced in building and maintaining large-scale, end-to-end search systems, developing ML/DL models for NLP and speech recognition, and designing robust infrastructures using modern engineering principles.

AREA OF EXPERTISE

Python	Data Scientist	Software Engineering	Deep Learning	Database
Java	Machine Learning	Backend Engineering	Spring Framework	AWS Cloud

WORK EXPERIENCE

Sr. Machine Learning Engineer , Insider **Aug 2022 - Present**

- Designed and implemented an end-to-end search engine architecture, integrating scalable infrastructure, modern software engineering practices, and robust API ecosystems.
- Developed semantic search capabilities using vector-based retrieval techniques and language model embeddings to improve relevance and user experience.
- Optimized and maintained Elasticsearch clusters, ensuring high availability and low-latency performance for large-scale production environments.
- Led the development of a merchandising sub-product with a focus on personalized ranking, filtering, and product discovery.
- Mentored junior engineers and collaborated with cross-functional teams to deliver ML-driven solutions aligned with product goals.

Data Scientist, Getir **May 2022 - July 2022**

- Analyzed large-scale datasets to extract insights and identify patterns relevant to NLP applications.
- Developed and fine-tuned machine learning and deep learning models for natural language processing tasks, including text classification and intent recognition.
- Prepared models for production deployment, ensuring scalability, efficiency, and alignment with business requirements.

Research Enginner, AdresGezgini **July 2020 - May 2022**

- Conducted research and develop on cutting-edge ML and DL methodologies for Turkish speech recognition and natural language processing tasks.
- Developed, trained, and fine-tuned models using TensorFlow, PyTorch, and Hugging Face Transformers for tasks such as automatic speech recognition (ASR), text classification, and named entity recognition.
- Deployed models across various platforms, including cloud-based environments and on-premise systems, optimizing for latency and scalability.
- Led experimentation and benchmarking efforts, improving model accuracy and performance through data augmentation and hyperparameter tuning.

EDUCATION

Master of Science in Electrical and Electronics Engineering **2021 - Discontinued**
Izmir Democracy University

- Specialized in speech recognition, with a focus on developing deep learning models for Turkish automatic speech recognition (ASR).
- Conducted research on acoustic modeling, language modeling, and end-to-end ASR systems using modern frameworks such as TensorFlow and PyTorch.

Bachelor of Science in Computer Engineering **2017 - 2021**
Kütahya Dumlupınar University

- GPA : 3.63

Bachelor of Science in Computer Engineering **2019 - 2020**
Opole University of Technology (ERASMUS)

- GPA : 3.65

SCIENTIFIC PUBLICATIONS

- **Journals:**
  - "Detection of Negative Cals in Call Centers with Convolutional Neural Networks", Journal of Information Technologies, 2023.
  - "Deep Feature Generation for Author Identification", Celal Bayar University Journal of Science, 2021.
  - “Increasing the Clasification Acuracy of BERT Model with Zero-Shot Learning”,TBV Journal of Computer Science and Engineering, 2021.
- **Conference Papers:**
  - “Increasing Performance in Turkish by Finetuning of Multilingual Speech to Text Model”, The 30th IEEE Conference on Signal Processing and Communications Applications (SIU), 2022.
  - "Auto-tagging of Short Conversational Sentences using Transformer Methods", 2021 Innovations in Inteligent Systems and Applications Conference (ASYU), 2021.
  - "Ad Text Clasification with Bidirectional Encoder Representations", 2021 6th International Conference on Computer Science and Engineering (UBMK), 2021.

ADDITIONAL INFORMATION

- **Languages:** English, Turkish
- **Certifications:**
  - AWS Certified Machine Learning – Specialty
  - AWS Certified Developer – Associate
  - AWS Certified Data Analytics – Specialty
  - AWS Certified Solutions Architect – Associate
  - Data Science Methodology
  - Python for Data Science, AI & Development
- **Projects:**
  - Search Engine Application,
  - General Purpose Chatbot Application
  - Automatic Detection of Negative Calls
  - Speech-to-Text for Turkish Language